Multiplying Polynomials

Concept

Multiply every term in one polynomial by every term in the other.

Distribute

$(a + b)(c + d) = ac + ad + bc + bd$.

Square

$(a \pm b)^2 = a^2 \pm 2ab + b^2$.

Difference of squares

$(a + b)(a - b) = a^2 - b^2$.

Check

Count products, preserve signs, then collect like terms.

Example 1

Expand $(2x - 3)(x^2 + x + 5)$.

Solution

1. Distribute: $2x^3 + 2x^2 + 10x - 3x^2 - 3x - 15$.

2. Collect terms: $2x^3 - x^2 + 7x - 15$.

Multiplying Polynomials

Example 2

Expand $(3x - 2)^2$.

Solution

1. Use $a^2 - 2ab + b^2$ with $a = 3x$, $b = 2$.

2. Result: $9x^2 - 12x + 4$.

Multiplying Polynomials

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Expand $(x + 6)(x + 2)$.

2. Expand $(2x + 1)(3x - 4)$.

3. Expand $(x - 5)^2$.

4. Expand $(4x + 3)(4x - 3)$.

5. Expand $(x + 2)(x^2 - x + 3)$.

Multiplying Polynomials

Practice answers | check after trying

1. $x^2 + 8x + 12$.

2. $6x^2 - 8x + 3x - 4 = 6x^2 - 5x - 4$.

3. $x^2 - 10x + 25$.

4. $16x^2 - 9$.

5. $x^3 - x^2 + 3x + 2x^2 - 2x + 6 = x^3 + x^2 + x + 6$.